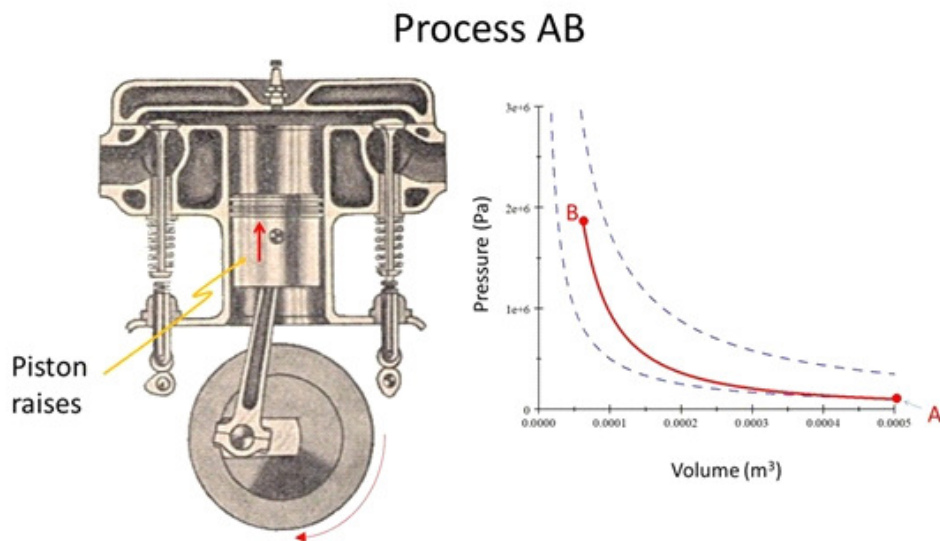


1.1 Process $A \rightarrow B$



Assignment:

Fill in the table for process $A \rightarrow B$

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
A	293	100	0.000 5	2.0525×10^{-2}		—
B			6.25×10^{-5}	2.0525×10^{-2}	287.17	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
AB						

ANSWER:

Process $A \rightarrow B$ is an adiabatic compression. The piston moves upward, compressing the fuel-air mixture. This happens rapidly, so there is no time for energy to leave by heat, $Q = 0$. That makes it close enough to being adiabatic that we can say this process is approximately adiabatic. In this process $\Delta E_{int} = w_{int}$.

We can use one of our adiabatic equations

$$P_i V_i^\gamma = P_f V_f^\gamma$$

to find the final pressure for this process at B

$$P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$$

But of course this works for any point on the line from A to B so

$$P = P_A \frac{V_A^\gamma}{V^\gamma}$$

We can use this form for P to find the work for process $A \rightarrow B$

$$\begin{aligned} w_{AB} &= - \int_{V_A}^{V_B} P dV \\ &= - \int_{V_A}^{V_B} P_A \frac{V_A^\gamma}{V^\gamma} dV \\ &= -P_A V_A^\gamma \int_{V_A}^{V_B} \frac{1}{V^\gamma} dV \\ &= -P_A V_A^\gamma \left(\frac{V^{1-\gamma}}{1-\gamma} \right) \Big|_{V_A}^{V_B} \\ &= -\frac{P_A V_A^\gamma}{1-\gamma} (V_B^{1-\gamma} - V_A^{1-\gamma}) \end{aligned}$$

We can rearrange this into a more convenient formula

$$\begin{aligned} w_{AB} &= \frac{P_A V_A^\gamma}{\gamma-1} \left(\frac{1}{V_B^{\gamma-1}} - \frac{1}{V_A^{\gamma-1}} \right) \\ &= \frac{1}{\gamma-1} \left(\frac{P_A V_A^\gamma}{V_B^{\gamma-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right) \\ &= \frac{1}{\gamma-1} \left(\frac{P_A V_A^\gamma}{V_B^\gamma V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right) \end{aligned}$$

And once again using

$$P_B V_B^\gamma = P_A V_A^\gamma$$

or

$$P_A = P_B \frac{V_B^\gamma}{V_A^\gamma}$$

We get

$$w_{AB} = \frac{1}{\gamma-1} \left(\frac{P_B \frac{V_B^\gamma}{V_A^\gamma} V_A^\gamma}{V_B^\gamma V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right)$$

A lot of things cancel. We get

$$\begin{aligned} w_{AB} &= \frac{1}{\gamma-1} \left(\frac{P_B}{V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right) \\ &= \frac{1}{\gamma-1} (P_B V_B - P_A V_A) \end{aligned}$$

which we can generalize into yet another basic equation for adiabatic processes!

$$w_{if} = \left(\frac{1}{\gamma - 1} \right) (P_f V_f - P_i V_i)$$

Then for process $A \rightarrow B$ we have

$$w_{AB} = \left(\frac{1}{\gamma - 1} \right) (P_B V_B - P_A V_A)$$

and we can use our equation for P_B that we got before, $P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$ so that our w_{AB} equation becomes

$$w_{AB} = \left(\frac{1}{\gamma - 1} \right) \left(P_A \frac{V_A^\gamma}{V_B^\gamma} V_B - P_A V_A \right)$$

And from our compression ratio equation

$$V_B = \frac{V_A}{8.00}$$

so w_{AB} becomes

$$\begin{aligned} w_{AB} &= \left(\frac{1}{\gamma - 1} \right) \left(P_A \frac{V_A^\gamma}{\left(\frac{V_A}{8} \right)^\gamma} \frac{V_A}{8.00} - P_A V_A \right) \\ &= \left(\frac{1}{\gamma - 1} \right) \left(8^\gamma P_A \left(\frac{V_A}{8} \right) - P_A V_A \right) \\ &= \left(\frac{V_A}{\gamma - 1} \right) (8^{\gamma-1} P_A - P_A) \\ &= \left(\frac{V_A P_A}{\gamma - 1} \right) (8^{\gamma-1} - 1) \end{aligned}$$

So the work is

$$w_{AB} = \left(\frac{V_A P_A}{\gamma - 1} \right) (8^{\gamma-1} - 1)$$

and we know all of these numbers so

$$\begin{aligned} w_{AB} &= \left(\frac{(0.0005 \text{ m}^3)(100 \text{ kPa})}{1.4 - 1} \right) (8^{1.4-1} - 1) \\ &= 162.17 \text{ J} \end{aligned}$$

But this is the work done on the gas, the useful work is the negative of this

$$w_{engAB} = -162.17 \text{ J}$$

Which tells us that it took energy to make the process $A \rightarrow B$ happen. You might need a pull cord to start this engine or have an electric starter motor. The change in internal energy is

$$\Delta E_{AB} = 162.17 \text{ J}$$

and we can find

$$E_{int} = \frac{5}{2}nRT$$

but we need n . We can get it from what we know at A using the ideal gas law

$$\begin{aligned} n &= \frac{P_A V_A}{RT_A} \\ &= \frac{(100 \text{ kPa}) (0.0005 \text{ m}^3)}{\left(8.314 \frac{\text{J}}{\text{mol K}}\right) (293 \text{ K})} \\ &= 2.0525 \times 10^{-2} \text{ mol} \end{aligned}$$

Then

$$\begin{aligned} E_A &= \frac{5}{2}nRT_A \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (293 \text{ K}) \\ &= 125.00 \text{ J} \end{aligned}$$

We used

$$P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$$

but we should find a numeric answer for P_B . Let's use our compression ratio equation for V_B again

$$\begin{aligned} P_B &= P_A \frac{V_A^\gamma}{\left(\frac{V_A}{8}\right)^\gamma} \\ &= P_A \frac{V_A^\gamma}{\frac{V_A^\gamma}{8^\gamma}} \\ &= 8^\gamma P_A \\ &= 8^{1.4} (100 \text{ kPa}) \\ &= 1837.9 \text{ kPa} \end{aligned}$$

And finally, from the ideal gas law

$$\begin{aligned} T_B &= \frac{P_B V_B}{nR} \\ &= \frac{(1837.9 \text{ kPa}) (6.25 \times 10^{-5} \text{ m}^3)}{(2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right)} \\ &= 673.15 \text{ K} \end{aligned}$$

At last we can find E_B

$$\begin{aligned} E_B &= \frac{5}{2}nRT_B \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (673.15 \text{ K}) \\ &= 287.17 \text{ J} \end{aligned}$$

what we have learned is summarized in this table

State	T (K)	P (kPa)	V (m ³)	n (mol)	E_{int} (J)	—
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00	—
B	673.15	1837.9	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	287.17	—
Process	Q (J)	w_{int} (J)	ΔE_{int} (J)	w_{eng} (J)	Q_h	Q_c
AB	0	162.17 J	162.17 J	−162.17 J	0	0

That was just one process!